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# FLOOR & CEILING

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# 1. FLOOR, CEILING AND FRACTIONAL FUNCTIONS

## 1.1 Basics

### A. Floor Operator

#### 1.1: Floor / Greatest Integer Operator

$$\lfloor x \rfloor = \text{The largest integer } \leq x$$

*Notation*

- The floor operator “**rounds down**” a number. It gets rid of any value after the decimal point.
- On the number line, the floor function finds the first integer that we encounter if go left from  $x$ . If the starting point is itself an integer, then we encounter  $x$ .

#### Example 1.2

Evaluate:

- A.  $\lfloor 5.6 \rfloor$
- B.  $\left\lfloor \frac{91}{6} \right\rfloor$
- C.  $\lfloor \pi \rfloor$

$$\lfloor 5.6 \rfloor = 5$$



$$\left\lfloor \frac{91}{6} \right\rfloor = \left\lfloor 15 \frac{1}{6} \right\rfloor = 15$$
$$\lfloor \pi \rfloor = \lfloor \approx 3.14 \rfloor = 3$$

#### 1.3: Floor Operator for Integers

For any integer  $x$ :

$$\lfloor x \rfloor = x$$

The floor operator never changes the value of an integer.

#### 1.4: Floor Operator for Negative Numbers

The floor operator rounds a number “down”.

It always makes the smaller, which means the output of the floor operator will be at the same place, or to the left of the input, on the number line.

It will be never be on the right.

#### Example 1.5

Evaluate:

- A.  $\left\lfloor \frac{81}{27} \right\rfloor$
- B.  $\lfloor -99 \rfloor$

$$\left\lfloor \frac{81}{27} \right\rfloor = \lfloor 3 \rfloor = 3$$

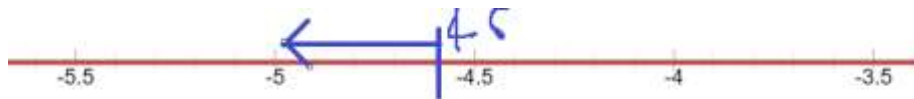
$$\lfloor -99 \rfloor = -99$$

### Example 1.6

Evaluate:

- A.  $\lfloor -4.6 \rfloor$
- B.  $\lfloor -9.99 \rfloor$
- C.  $\left\lfloor -\frac{12}{5} \right\rfloor$
- D.  $\left\lfloor -\frac{89}{7} \right\rfloor$
- E.  $\lfloor -\pi \rfloor, \pi \approx 3.14$
- F.  $\lfloor -2\pi \rfloor$
- G.  $\left\lfloor -\frac{\pi}{2} \right\rfloor$
- H.  $\lfloor -e \rfloor, e \approx 2.71$
- I.  $\lfloor -2e \rfloor$
- J.  $\left\lfloor -\frac{e}{2} \right\rfloor$
- K.  $\left\lfloor -\frac{1002}{11} \right\rfloor$

$$\lfloor -4.6 \rfloor = -5$$



$$\lfloor -9.99 \rfloor = -10$$

$$\left\lfloor -\frac{12}{5} \right\rfloor = \left\lfloor -2\frac{2}{5} \right\rfloor = \lfloor -3 \rfloor$$

$$\left\lfloor -\frac{89}{7} \right\rfloor = \left\lfloor -12\frac{5}{7} \right\rfloor = -13$$

$$\lfloor -\pi \rfloor = \lfloor \approx -3.14 \rfloor = -4$$

$$\lfloor -2\pi \rfloor = \lfloor \approx -6.28 \rfloor = -7$$

$$\left\lfloor -\frac{\pi}{2} \right\rfloor = \left\lfloor \approx -\frac{3.14}{2} \right\rfloor = \lfloor \approx -1.57 \rfloor = -2$$

$$\lfloor -e \rfloor = \lfloor \approx -2.71 \rfloor = -3$$

$$\lfloor -2e \rfloor = \lfloor \approx -5.42 \rfloor = -6$$

### Example 1.7

The greatest integer function of  $x$  is expressed as  $\lfloor x \rfloor$  and is defined to be the greatest integer less than or equal to  $x$ . Find  $\lfloor \pi - 4 \rfloor$  (**MathCounts 1995 Warm-Up 7**)

$$\lfloor \pi - 4 \rfloor = \lfloor \approx -0.86 \rfloor = -1$$

## B. Real Life Application

### Example 1.8

A packet of biscuits costs Rs. 10. How many packets can the following people buy, and how much money will they have left over?

- A. Dave has 73 Rs.
- B. William has 89 Rs and 50 Paise.

Dave can buy 7 packets for:

$$7 \times 10 = 70 \text{ Rs} \Rightarrow \text{Money left over} = \text{Rs. } 3$$

Dave can buy 8 packets for:

$$8 \times 10 = 80 \text{ Rs} \Rightarrow \text{Money left over} = \text{Rs. } 9.50$$

The above question was solved using logic, and the realization that the number of packets bought must be a natural number.

The same logic can be expressed in terms of the floor function. Do that.

One packet costs Rs. 10. Hence, if you have Rs.  $x$ , the number of packets that you can buy should be:

$$\frac{x}{10}$$

However,  $\frac{x}{10}$  may not be an integer, and we want to round down, hence, we apply the floor operator, giving us:

$$\text{No. of Packets that can be bought} = \left\lfloor \frac{x}{10} \right\rfloor$$

$$\text{No. of Packets for Dave} = \left\lfloor \frac{x}{10} \right\rfloor = \left\lfloor \frac{73}{10} \right\rfloor = \lfloor 7.3 \rfloor = 7$$

$$\text{No. of Packets for William} = \left\lfloor \frac{89.5}{10} \right\rfloor = \lfloor 8.95 \rfloor = 8$$

## C. Floor Function

### Example 1.9

Does the floor operator represent a function? That is, is

$$y = \lfloor x \rfloor$$

A function?

Yes, because every input has exactly one output.

### Example 1.10

Consider the function

$$y = \lfloor x \rfloor$$

What are its

- A. Domain (Recall that domain is valid  $x$  values.)
- B. Range (Recall that range is valid  $y$  values.)

$$\begin{aligned} \text{Domain} &= \mathbb{R} = (-\infty, \infty) \\ \text{Range} &= \mathbb{Z} = \text{All Integers} \end{aligned}$$

### Example 1.11

The graph of the floor function is also called the stair function. (*Think why?*) Graph

$$f(x) = \lfloor x \rfloor$$

- A. Does it pass the vertical line test? (The vertical line test is a test that says that if no vertical line intersects a graph more than once, then the graph is a function).
- B. What is the significance?

#### Vertical Line Test

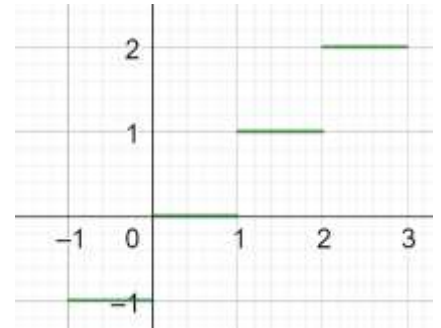
Yes, any vertical line intersects the graph of the function at most once.  
Note that at integers, there should be a “hole”, which is not shown in the graph.

For example, every line should have

*an empty circle at the right endpoint*  
*a filled circle at the left endpoint*

#### Significance

The vertical line test is the graphical test for whether a graph is a function.  
This confirms that the floor operator is a function.



- C. Does it pass the horizontal line test? (The horizontal line test is a test that says that if no horizontal line intersects a function more than once, then the function is an invertible function).
- D. What is the significance?

#### Horizontal Line Test

If you draw a horizontal line at

$$y = 0, 1, 2, 3, \dots$$

It intersects the graph of the function in an infinite number of places.

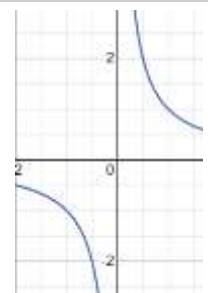
Hence, the graph fails the horizontal line test.

#### Significance

The horizontal line test is the graphical test for whether a function has an inverse. Since the test fails, the floor function does not have an inverse.

### 1.12: Continuous Function

A continuous function is one that can be drawn without lifting your pencil.



### 1.13: Types of Discontinuity

There are three types of discontinuity:

- Jump Discontinuity: The graph jumps from one value to another.
- Infinite Discontinuity: The graph becomes infinite next to a value, and is not defined at that value.
- Removable Discontinuity: The graph has a hole.

### Example 1.14

The graph of

$$f(x) = \lfloor x \rfloor$$

Is

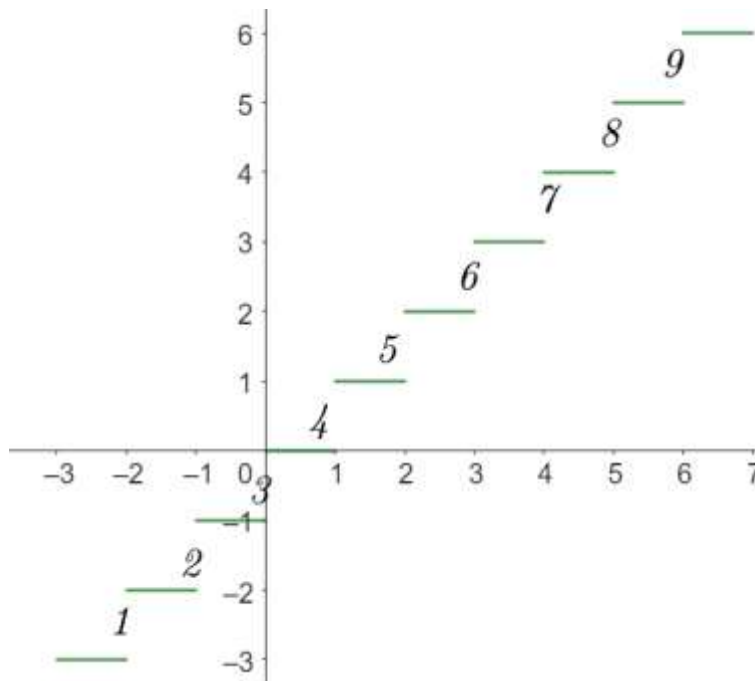
- A. Continuous
- B. Not continuous with an infinite discontinuity
- C. Not continuous with a jump discontinuity
- D. Not continuous with a removable discontinuity

### Example 1.15

$$f(x) = [x], \quad -3 \leq x < 7$$

What is the number of jump discontinuities that the graph of the above function has?

$$\{-2, -1, 0, 1, 2, 3, 4, 5, 6\} \Rightarrow 6 - (-2) + 1 = 6 + 2 + 1 = 9$$



### Example 1.16

The floor function can be defined piece-wise. Consider:

$$f(x) = [x], \quad D_f = [0, 3)$$

- A. Graph the function.
- B. Write this as a piece-wise function.

$$f(x) = \begin{cases} 0, & 0 \leq x < 1 \\ 1, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \end{cases}$$

Extend the piece-wise definition to include all real numbers.

$$f(x) = n, \quad n \leq x < n + 1, n \in \mathbb{Z}$$

### Example 1.17

Let

$$f(x) = [x]$$

$f^n(x)$  is the function applied  $n$  times

Find

$$f^{2021}(2020.2019)$$

Build up the logic

$$f(2020.2019) = 2020$$

$$f^2(2020.2019) = f(2020) = 2020$$

$$f^3(2020.2019) = f^2(2020) = f(2020) = 2020$$

And from the above, we can see the pattern.

Hence, we can conclude that:

$$f^{2021}(2020.2019) = f^{2020}(2020) = 2020$$

### Example 1.18

Note: For this question, we need the range of  $x$ , which refers to the set of values that  $x$  can take. Do not confuse it with the range of  $f(x)$ .

$$\text{Floor Function} = f(x) = \lfloor x \rfloor$$

Determine the range that  $x$  can take if:

- A.  $f(x) = 3$
- B.  $f(x) = n$

$$[3, 4)$$

$$[n, n + 1)$$

## D. Ceiling Operator

### 1.19: Ceiling Operator / Least Integer Operator

$$\lceil x \rceil = \text{The smallest integer } \geq x$$

*Notation*

- The ceiling function is very similar to the floor function in terms of its idea.
- Instead of going to the left on the number line, we go to the right on the number line, or stay at the same position.

In other words, the floor function “**rounds up**” a number.

### Example 1.20

Evaluate:

- A.  $\lceil 5.6 \rceil$
- B.  $\left\lceil \frac{91}{6} \right\rceil$
- C.  $\lceil e \rceil$ ,  $e \approx 2.71$

$$\lceil 5.6 \rceil = 6$$

$$\left\lceil \frac{91}{6} \right\rceil = \left\lceil 15 \frac{1}{6} \right\rceil = 16$$

$$\lceil e \rceil = \lceil \approx 2.71 \rceil = 3$$

### Example 1.21

Evaluate:

- A.  $\lceil 23 \rceil$



B.  $[-54]$

$$\begin{aligned}[23] &= 23 \\ [-54] &= -54\end{aligned}$$

### Example 1.22

Evaluate:

- A.  $[-5.6]$
- B.  $\left[-\frac{89}{7}\right]$
- C.  $[-e]$ ,  $e \approx 2.71$

$$\begin{aligned}[-5.6] &= -5 \\ \left[-\frac{89}{7}\right] &= \left[-12\frac{5}{7}\right] = -12 \\ [-e] &= [\approx -2.71] = -2\end{aligned}$$

### Example 1.23

I have a party, and each person present is expected to eat 3 slices of pizza. Each pizza has 4 slices. I have 2 friends coming over (and I will also be eating). What is the number of pizzas that I should order?

$$\text{No. of Slices} = 3 \times 3 = 9$$

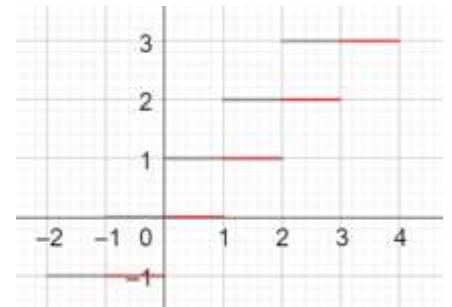
$$\text{No. of Pizzas} = \frac{9}{4} = 2\frac{1}{4} \Rightarrow 3 \text{ Pizzas}$$

$$\text{No. of Pizzas} = \left\lceil \frac{\text{Slices per person} \times \text{No. of People}}{\text{Slices per Pizza}} \right\rceil = \left\lceil \frac{3 \times 3}{4} \right\rceil = \left\lceil \frac{9}{4} \right\rceil = \left\lceil 2\frac{1}{4} \right\rceil = 3$$

### 1.24: Floor and Ceiling for Integers

$$[x] = [x] \Leftrightarrow x \in \mathbb{Z}$$

- Graphically, the only point of intersection of the graphs  
 $y = [x], y = \lceil x \rceil$   
 Is at the integers
- If the floor of a number has the same value as the ceiling of a number, then the number must be an integer. It cannot have a decimal part which is non-zero, since then the floor function will have an output which is one number less than the output of the ceiling function.



### Example 1.25

What kind of number is  $x$  if:

$$[x] = \lceil x \rceil = x$$

$$\text{Consider } 4.2 \Rightarrow [4.2] = 4, \lceil 4.2 \rceil = 5$$

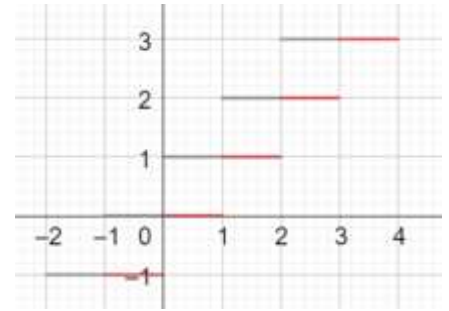
$\therefore$  Any number with a value after the decimal point will not work.

$$\therefore \{x\} = 0 \Rightarrow x \in \mathbb{Z}$$

## 1.26: Floor and Ceiling for non-Integral Values

$$\lceil x \rceil - \lfloor x \rfloor = 1, \quad \underbrace{x \in \mathbb{R} - \{\mathbb{Z}\}}_{\substack{x \text{ is a fraction,} \\ \text{but not an integer}}}$$

If  $x$  is not an integer, then the difference between  $\lfloor x \rfloor$ , and ceiling of  $\lceil x \rceil$  is one.



### Example 1.27

Let

$$\begin{aligned} f(x) &= \lfloor x \rfloor, & \text{floor function} \\ g(x) &= \lceil x \rceil, & \text{ceiling function} \end{aligned}$$

Evaluate:

- $f(x) - g(x)$ , if  $x$  is a fraction
- $f(x) - g(x)$ , if  $x$  is an integer
- $3g(x) - 2f(x)$ , if  $x$  is a fraction

$$f(x) - g(x) = \text{Floor} - \text{Ceiling} = -1$$

$$f(x) - g(x) = 0$$

$$3g(x) - 2f(x) = g(x) + 2[g(x) - f(x)] = \lceil x \rceil + 2$$

## E. Function, Graph, Domain and Range

We have already seen the graph of the floor function, and analyzed its domain and range. This is similar for the ceiling function, and hence we see it in one go.

### Example 1.28

Graph

$$y = \lceil x \rceil$$

- Show that it is a function, algebraically.
- Show that it is a function, graphically.
- What is the domain and range of the function?
- Compare the graph of the ceiling function with that of the floor function.
- Discuss the continuity of the ceiling function.

#### Part A

Every input to  $\lceil x \rceil$  has precisely one output.

Hence, it is a function.

#### Part B

Any vertical line will intersect the graph in a maximum of one place.

Hence, it is a function.

#### Part C

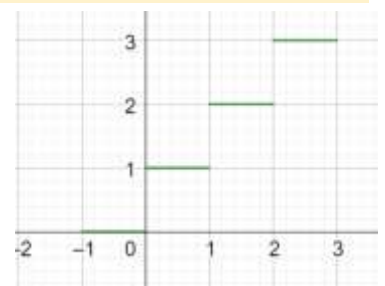
$$\text{Domain} = \mathbb{R}$$

$$\text{Range} = \mathbb{Z}$$

#### Part D

$$\lceil x \rceil = \lfloor x \rfloor + 1$$

Hence, the graph of the ceiling function is the same as the graph of the floor function, moved up by 1 unit.



### Example 1.29

Note: For this question, we need the range of  $x$ , which refers to the set of values that  $x$  can take. Do not confuse it with the range of  $f(x)$ .

$$f(x) = \lceil x \rceil$$

Determine the range that  $x$  can take if:

- A.  $f(x) = 3$
- B.  $f(x) = n$

$$(2, 3]$$

$$(n - 1, n]$$

## F. Fractional Part Operator

### 1.30: Fractional Part Operator

$\{x\}$  = The fractional part of  $x = x - [x], x > 0$

*Notation*

$\{x\}$  = The fractional part of  $x = x - [x], x < 0$

*Notation*

#### Example 1.31

Evaluate the below using the definition of fractional part function

- A.  $\{5.6\}$
- B.  $\left\{\frac{91}{6}\right\}$

$$\{5.6\} = 5.6 - [5.6] = 5.6 - 5 = 0.6$$

#### Example 1.32

Evaluate the below using the definition of fractional part function

- A.  $\{-5.6\}$
- B.  $\left\{-\frac{89}{7}\right\}$

$$\{-5.6\} = -5.6 - [-5.6] = -5.6 - (-5) = -5.6 + 5 = -0.6$$

### 1.33: Fractional Part of Irrational Numbers

- An irrational number does not have a fractional part that can be written as a fraction.
- Hence, the fractional part of an irrational number is an irrational number (not a fraction).

$$\underbrace{\pi}_{\text{Irrational Number}} - \underbrace{3}_{\text{Rational Number}} = \underbrace{0.\text{Something}}_{\text{Irrational Number}}$$

#### Example 1.34

Arshi does the following calculation. Comment on whether she is correct.

$$\{\pi\} = \left\{\frac{22}{7}\right\} = \left\{3\frac{1}{7}\right\} = \frac{1}{7}$$

$$\pi \text{ is irrational} \Rightarrow \{\pi\} \text{ is also irrational}$$

But

$$\frac{1}{7} \text{ is rational} \Rightarrow \text{Contradiction}$$

The contradiction happened because the calculation was wrong.

Hence, the above working by Arshi is not correct.

### 1.35: Rational Number

A rational number is a number that can be written

- In the form of a fraction
- Where the numerator and the denominator are integers
- And the denominator is not zero

$$\mathbb{Q} = \left\{ x \mid x = \frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q \neq 0 \right\}$$

$\pi$  is irrational  $\Rightarrow$  It cannot be written as  $\frac{p}{q}, p, q \in \mathbb{Z} \Rightarrow \pi \approx \frac{22}{7} \Rightarrow \{\pi\} \neq \left\{\frac{22}{7}\right\}$

$$\frac{1}{7} = 0.142857 \Rightarrow \text{Non-terminating, repeating decimal}$$

$\pi \Rightarrow \text{Non-terminating, non-repeating decimal}$

## G. Fractional Part Function

### Example 1.36

Graph

$$y = \{x\}$$

- A. Show that it is a function, algebraically.
- B. Show that it is a function, graphically.
- C. What is the domain and range of the function?
- D. Compare the graph of the fractional part function with that of the floor function.
- E. Discuss the continuity of the fractional part function.

#### Part A

For every input,  $\{x\}$  has exactly one output. Hence, it is a function.

#### Part B

The fractional part graph is drawn alongside.

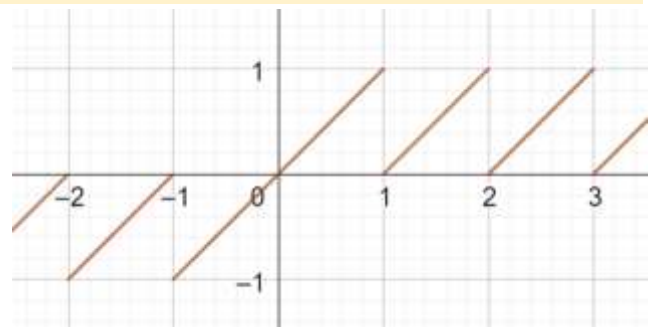
Any vertical line crosses the graph in exactly one place.

Hence, the graph passes the vertical line test, and hence it is a function.

#### Part C

$$\text{Domain} = \mathbb{R} = (-\infty, \infty)$$

$$\text{Range} = (-1, 1)$$



### Example 1.37

- A. Graph the floor function and the fractional part function on the same axes. In how many places do the graph of the floor function and the fractional part function intersect?
- B. What does this tell you about the solution to  $\lfloor x \rfloor = \{x\}$

The only place that the graphs intersect is when  
 $x = 0$

Because

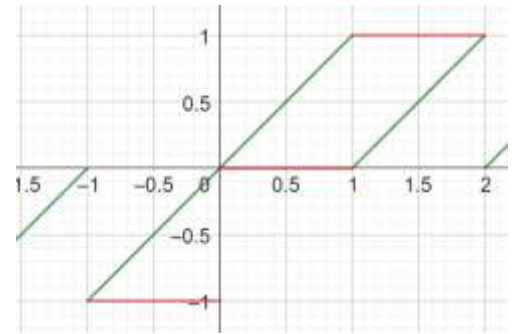
$$\lfloor 0 \rfloor = \{0\} = 0$$

$$x = 1 \Rightarrow \lfloor 1 \rfloor = 1, \{1\} = 0 \Rightarrow \text{No Intersection}$$

$$x = -1 \Rightarrow \lfloor -1 \rfloor = -1, \{-1\} = 0 \Rightarrow \text{No Intersection}$$

## H. Ceiling Function (Alternate Definition)

We can define the ceiling function in terms of the fractional function.



### 1.38: Ceiling and Fractional Part

$$\lceil x \rceil = x - \{x\} + 1$$

#### Example 1.39

Verify that  $\lceil x \rceil = x - \{x\} + 1$  holds for:

- A. 7.2
- B.  $\frac{71}{9}$
- C.  $-\frac{72}{11}$

## I. Mixed Review

#### Example 1.40

$$X = \left\{3.4, \pi, e, \frac{99}{100}, 1\right\}$$

Let

$Y$  be the sum of the result of applying the floor function to each number in  $X$ .

$Z$  be the sum of the result of applying the ceiling function to each number in  $X$ .

$P$  be the sum of the result of applying the fractional part function to each number in  $X$ .

Find the:

- A. product of  $Y$  and  $Z$
- B. positive difference between  $Y$  and  $Z$
- C. Find  $P$

$$Y = \lfloor 3.4 \rfloor + \lfloor \pi \rfloor + \lfloor e \rfloor + \left\lfloor \frac{99}{100} \right\rfloor + \lfloor 1 \rfloor = 3 + 3 + 2 + 0 + 1 = 9$$

$$Z = \lceil 3.4 \rceil + \lceil \pi \rceil + \lceil e \rceil + \left\lceil \frac{99}{100} \right\rceil + \lceil 1 \rceil = 4 + 4 + 3 + 1 + 1 = 13$$

$$YZ = 13 \times 9 = 117$$

$$Z - Y = \underbrace{\lfloor 3.4 \rfloor - \lceil 3.4 \rceil}_{-1} + \underbrace{\lfloor \pi \rfloor - \lceil \pi \rceil}_{-1} + \underbrace{\lfloor e \rfloor - \lceil e \rceil}_{-1} + \underbrace{\left\lfloor \frac{99}{100} \right\rfloor - \left\lceil \frac{99}{100} \right\rceil}_{-1} + \underbrace{\lfloor 1 \rfloor - \lceil 1 \rceil}_{0} = -1 - 1 - 1 - 1 + 0 = -4$$

$$P = \underbrace{\{3.4\}}_{0.4} + \underbrace{\left\{\frac{99}{100}\right\}}_{0.99} + \underbrace{\{1\}}_{0} + \underbrace{\left\{\frac{22}{7}\right\}}_{\frac{1}{7}} = 0.4 + 0.99 + 0 + \frac{1}{7}$$

## J. Expressions

### Example 1.41

Simplify:

- A.  $\lceil 3.43 \rceil - \lfloor 3.43 \rfloor$
- B.  $\lfloor -7.12 \rfloor - \lceil -7.12 \rceil$

### 1.42: Floor and Ceiling

$$\lceil x \rceil - \lfloor x \rfloor = 1, x \notin \mathbb{Z}$$

$$\lceil x \rceil - \lfloor x \rfloor = 0, x \in \mathbb{Z}$$

### Example 1.43

Simplify:

$$2\lfloor x \rfloor - 3\lceil x \rceil$$

(Ensure you cover all possible cases)

Case I:  $x$  is not an integer

$$2\lfloor x \rfloor - 3\lceil x \rceil = 2(\lfloor x \rfloor - \lceil x \rceil) - \lceil x \rceil = -2 - \lceil x \rceil$$

Case II:  $x$  is an integer

$$2\lfloor x \rfloor - 3\lceil x \rceil = 2(\lfloor x \rfloor - \lceil x \rceil) - \lceil x \rceil = -\lceil x \rceil$$

### Example 1.44

Absolute Value

### Example 1.45

Evaluate:

- A.  $\lfloor 3.1^2 \rfloor - (\lfloor 3.1 \rfloor)^2$
- B.  $\lfloor 2.5^2 \rfloor - (\lfloor 2.5 \rfloor)^2$

$$\lfloor 3.1^2 \rfloor - \lfloor 3.1 \rfloor \lfloor 3.1 \rfloor = \lfloor 9.61 \rfloor - 3 \times 3 = 9 - 9 = 0$$

$$\lfloor 2.5^2 \rfloor - \lfloor 2.5 \rfloor \lfloor 2.5 \rfloor = \lfloor 6.25 \rfloor - 2(2) = 6 - 4 = 2$$

### 1.46: Interchanging Operations

In general, the floor/ceiling functions cannot be interchanged with squares and square roots.

$$\lfloor x^2 \rfloor \neq \lfloor x \rfloor \cdot \lfloor x \rfloor$$

## K. Square Roots

### Example 1.47

## L. Nested Expressions

### Example 1.48

## 1.2 Transformations

### A. Horizontal Shift

#### Example 1.49

Graph  $y = \lfloor x \rfloor$ . Then solve the equations. Make sure you are able to see the connection between the graph and the equations.

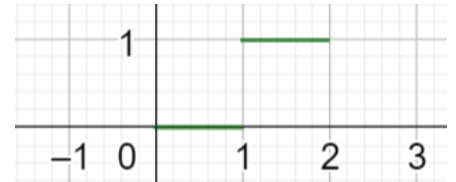
- A.  $\lfloor x \rfloor = 0$
- B.  $\lfloor x \rfloor = 1$

#### Part A

From the graph, we can see that:

$$\lfloor x \rfloor = 0 \Rightarrow 0 \leq x < 1$$

We need to be careful about the endpoints. The left endpoint is included, but the right is not.



#### Part B

$$x = 1 \Rightarrow 1 \leq x < 2$$

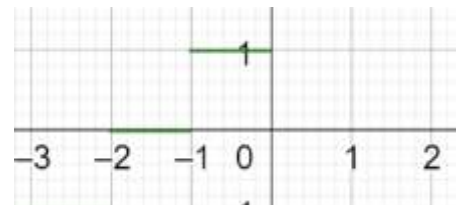
#### Example 1.50

Evaluate  $f(x) = \lfloor x + 2 \rfloor$  for

- A.  $x = 0.5$
- B.  $x = -0.5$

$$x = 0.5 \Rightarrow \lfloor x + 2 \rfloor = \lfloor 0.5 + 2 \rfloor = \lfloor 2.5 \rfloor = 2$$

$$x = -0.5 \Rightarrow \lfloor x + 2 \rfloor = \lfloor -0.5 + 2 \rfloor = \lfloor 1.5 \rfloor = 1$$



#### Example 1.51

Graph  $y = \lfloor x + 2 \rfloor$ . Then, solve the following equations:

- A.  $\lfloor x + 2 \rfloor = 0$
- B.  $\lfloor x + 2 \rfloor = 1$

#### Part A

$$\lfloor x + 2 \rfloor = 0 \Rightarrow 0 \leq x + 2 < 1$$

Solving the inequality gives us:

$$-2 \leq x < -1 \Rightarrow x \in [-2, -1)$$

#### Part B

$$\lfloor x + 2 \rfloor = 1 \Rightarrow 1 \leq x + 2 < 2 \Rightarrow -1 \leq x < 0 \Rightarrow x \in [-1, 0)$$

#### Example 1.52

The adjoining diagram has, on the same axes:

$$y = \lfloor x \rfloor \Rightarrow \text{Red Graph}, \quad y = \lfloor x + 2 \rfloor \Rightarrow \text{Green Graph}$$

What is the relation between the two graphs?

The red graph is the same as the green graph shifted to the left by 2 units.



### 1.53: Horizontal Shift

$$y = \lfloor x + k \rfloor$$

$k > 0$ : Shifted left by  $k$  units  
 $k < 0$ : Shifted right by  $k$  units

Note the peculiarity here:  
 Adding  $k$  shifts the graph left (and not right).

### Example 1.54

Solve the following:

- A.  $\lfloor x + 5 \rfloor = 3$
- B.  $\lfloor x - 4 \rfloor = 7$
- C.  $\lfloor x + 9 \rfloor = -2$
- D.  $\lfloor x - 2 \rfloor + \frac{1}{3} = -4\frac{1}{3}$

## B. Vertical Shift

### 1.55: Vertical Shift

$y = \lfloor x \rfloor + k$   
 $k > 0$ : Shifted up by  $k$  units  
 $k < 0$ : Shifted down by  $k$  units

### Example 1.56

Graph  $y = \lfloor x \rfloor + k$ . Use that to solve:

A=

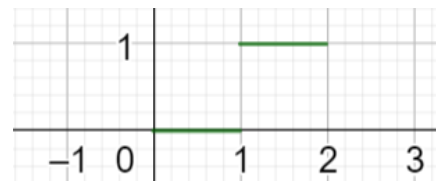
## C. No Solutions

### 1.57: No Solutions

The equation

$$\lfloor x \rfloor = n, n \notin \mathbb{Z}$$

Notice that the graph of  $y = \lfloor x \rfloor$  is a graph with jumps.  
 It has lines at the integers, but nothing in between.



### Example 1.58

Two equations are given below. One of them has a solution, while the other does not. Write the solution set for the one that does, and explain why the other does not.

- A.  $\lfloor x + 5 \rfloor = 3.5$
- B.  $\lfloor x + 5 \rfloor + 0.5 = 3.5$

### Example 1.59

Two equations are given below. One of them has a solution, while the other does not. Write the solution set for the one that does, and explain why the other does not.

- A.  $\lfloor x + 2 \rfloor + \frac{1}{3} = 3.\bar{3}$
- B.  $\lfloor x + 2 \rfloor + \frac{1}{3} = 3.33$



## D. Horizontal Stretch

### Example 1.60

Evaluate  $y = \left\lfloor \frac{x}{3} \right\rfloor$  when  $x = \{0, 1, 2, \dots, 10\}$

$$x = 0 \Rightarrow \left\lfloor \frac{x}{3} \right\rfloor = \left\lfloor \frac{0}{3} \right\rfloor = \lfloor 0 \rfloor = 0$$

$$x = 1 \Rightarrow \left\lfloor \frac{1}{3} \right\rfloor = \left\lfloor \frac{1}{3} \right\rfloor = 0$$

⋮

Summary:

$$x = \{0, 1, 2\} \Rightarrow \left\lfloor \frac{x}{3} \right\rfloor = 0$$

$$x = \{3, 4, 5\} \Rightarrow \left\lfloor \frac{x}{3} \right\rfloor = 1$$

$$x = \{6, 7, 8\} \Rightarrow \left\lfloor \frac{x}{3} \right\rfloor = 2$$

### Example 1.61

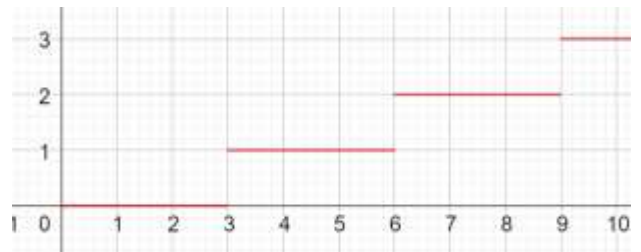
Graph  $y = \left\lfloor \frac{x}{3} \right\rfloor$ . Hence find the values that satisfy:

$$\left\lfloor \frac{x}{3} \right\rfloor = a, \quad a = \{1, 2, 3\}$$

$$\left\lfloor \frac{x}{3} \right\rfloor = 0 \Rightarrow 0 \leq x < 3$$

$$\left\lfloor \frac{x}{3} \right\rfloor = 1 \Rightarrow 3 \leq x < 6$$

$$\left\lfloor \frac{x}{3} \right\rfloor = 2 \Rightarrow 6 \leq x < 9$$



### 1.62: Horizontal Stretch

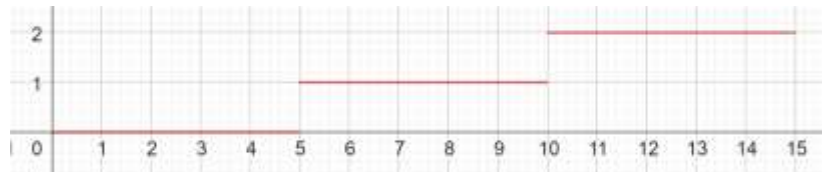
$$y = \left\lfloor \frac{x}{k} \right\rfloor$$

$k > 1 \Rightarrow$  Horizontal Stretch by a factor of  $k$

### Example 1.63

Use the graph of  $y = \left\lfloor \frac{x}{5} \right\rfloor$  to find the solution for:

$$\left\lfloor \frac{x}{5} \right\rfloor = a, \quad a = \{0, 1\}$$



$$\left\lfloor \frac{x}{5} \right\rfloor = 0 \Rightarrow 0 \leq x < 5 \Leftrightarrow [0, 5)$$

$$\left\lfloor \frac{x}{5} \right\rfloor = 1 \Rightarrow 5 \leq x < 10 \Leftrightarrow [5, 10)$$

## E. Horizontal Shrink

### 1.64: Horizontal Shrink

$$y = [kx]$$

$$k > 1 \Rightarrow \text{Horizontal Shrink by a factor of } k$$

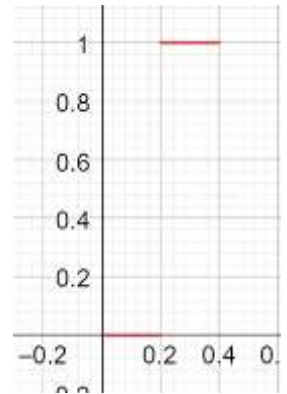
#### Example 1.65

Use the graph of  $y = [5x]$  to find the values that satisfy:

$$[5x] = a, a = \{0, 1\}$$

$$[5x] = 0 \Rightarrow 0 \leq x < \frac{1}{5}$$

$$[5x] = 1 \Rightarrow 0.2 \leq x < \frac{2}{5}$$



#### Example 1.66

Graph  $[3x]$ . Hence find the range of values that satisfies:

A.  $[3x] = 0$

B.  $[3x] = 1$

$$[3x] = 0 \Rightarrow 0 \leq x < \frac{1}{3}$$

$$[3x] = 1 \Rightarrow 0.2 \leq x < \frac{2}{3}$$

## F. Vertical Stretch/Shrink

### 1.67: Vertical Stretch/Shrink

$$y = k[x]$$

$$k > 1 \Rightarrow \text{Vertical Stretch}$$

$$0 < k < 1 \Rightarrow \text{Vertical Shrink}$$

#### Example 1.68

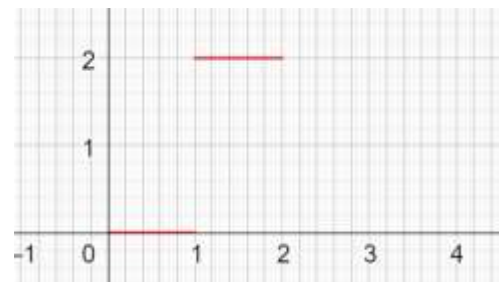
Graph  $y = 2[x]$  and use it to solve

$$2[x] = a, a = \{0, 1, 2\}$$

$$2[x] = 0 \Rightarrow [x] = 0 \Rightarrow x \in [0, 1)$$

$$2[x] = 1 \Rightarrow [x] = \frac{1}{2} \Rightarrow \text{Not Possible}$$

$$2[x] = 2 \Rightarrow [x] = \frac{2}{2} = 1$$



Solve

$$2[x] = n$$

$$2[x] = n \Rightarrow [x] = \frac{n}{2}$$

If

$$n \text{ is even, then } [x] = \frac{n}{2} \Rightarrow x \in \left[\frac{n}{2}, \frac{n}{2} + 1\right)$$

$n$  is not even  $\Rightarrow \frac{n}{2}$  is a fraction and hence there are no valid solutions

### Example 1.69

Solve:

- A.  $3[x] = 6$
- B.  $3[x] = 7$

$$3[x] = 6 \Rightarrow [x] = \frac{6}{3} = 2 \Rightarrow x \in [2, 3)$$
$$3[x] = 7 \Rightarrow [x] = \frac{7}{3} \Rightarrow \text{No Solutions} \Rightarrow x = \phi$$

## 1.3 Equations

### A. A Simple Property

#### 1.70: Simple Floor Equations

$$\text{For integer } n, [x] = n \Leftrightarrow n \leq x < n + 1$$

If  $n$  is not an integer, the equation has no solution.

### Example 1.71

Find the solution set that  $x$  can take.

- A.  $[x] = 0$
- B.  $[x] = 3$
- C.  $[x] = -27$
- D.  $[x] = 3.5$

$$[x] = 0 \Rightarrow 0 \leq x < 1 \Rightarrow x \in [0, 1)$$
$$[x] = 3 \Rightarrow 3 \leq x < 4 \Rightarrow x \in [3, 4)$$
$$[x] = -27 \Rightarrow -27 \leq x < -26 \Rightarrow x \in [-27, -26)$$
$$[x] = 3.5 \Rightarrow \text{No Solutions} \Rightarrow x \in \phi$$

#### 1.72: Simple Ceiling Equations

$$[x] = n \Leftrightarrow n - 1 \leq x < n$$

Note that  $n$  must be an integer, else the equation above has no solution.

### Example 1.73

Solve

- A.  $[x] = 0$
- B.  $[x] = 4$
- C.  $[x] = -3$
- D.  $[x] = 2.5$
- E.  $[x] = e, e \approx 2.71$
- F.  $[x] = 12$
- G.  $[x] = 5$
- H.  $[x] = \pi$
- I.  $2[x] + 1 = 3$

$$[x] = 0 \Rightarrow -1 < x \leq 0 \Rightarrow (-1, 0]$$

$$\begin{aligned}[x] = 4 &\Rightarrow 3 < x \leq 4 \Rightarrow (3, 4] \\ [x] = -3 &\Rightarrow -4 < x \leq -3 \Rightarrow (-4, -3] \\ [x] = 2.5 &\Rightarrow \text{No Solutions} \\ [x] = e &\Rightarrow [x] \approx 2.71 \Rightarrow \text{No Solutions} \\ [x] = 12 &\Rightarrow 11 < x \leq 12 \Rightarrow x \in (11, 12] \\ [x] = 5 &\Rightarrow 5 \leq x < 6 \Rightarrow x \in [5, 6) \\ [x] = \pi &\approx 3.14 \Rightarrow \text{No Solutions}\end{aligned}$$

### Example 1.74

Solve:

$$2[x] + 1 = 3$$

We will isolate the ceiling operator by “solving” for it, just like we solve for a variable.  
Subtract 1 from both sides:

$$2[x] = 2$$

Divide both sides by 2:

$$[x] = 1 \Rightarrow 0 < x \leq 1 \Rightarrow x \in (0, 1]$$

### 1.75: Connecting Ceiling with Floor

If  $x$  is not an integer, then the ceiling of  $x$  is always one more than the floor of  $x$ . That is:

$$x \notin \mathbb{Z} \Rightarrow [x] = \lfloor x \rfloor + 1$$

If  $x$  is an integer, then the floor and ceiling of  $x$  have the same value:

$$x \in \mathbb{Z} \Rightarrow [x] = \lfloor x \rfloor$$

### Example 1.76

Suppose

$$\lfloor x \rfloor = 34$$

Find  $\lceil x \rceil$  if:

- A.  $x$  is an integer
- B.  $x$  is not an integer

#### Part A

The only integer for which

$$\lfloor x \rfloor = 34$$

Is

$$x = 34 \Rightarrow \lceil x \rceil = 34$$

#### Part B

$$x \neq 34 \Rightarrow 34 < x < 35 \Rightarrow \lceil x \rceil = 35$$

## B. Addition and Subtraction

### Example 1.77

- A.  $\lfloor x + 3 \rfloor = 0$
- B.  $\lfloor x - 6 \rfloor = 0$
- C.  $\lfloor x + 3 \rfloor + 3 = 5$

#### Part A

$$\lfloor x + 3 \rfloor = 0 \Rightarrow 0 \leq x + 3 < 1 \Rightarrow -3 \leq x < -2 \Rightarrow x \in [-3, -2)$$

Part B

$$\lfloor x - 6 \rfloor = 0 \Rightarrow 0 \leq x - 6 < 1 \Rightarrow 6 \leq x < 7 \Rightarrow x \in [6, 7)$$

Part C

$$\lfloor x + 3 \rfloor = 2 \Rightarrow 2 \leq x + 3 < 3 \Rightarrow -1 \leq x < 0 \Rightarrow x \in [-1, 0)$$

### Example 1.78

- A.  $\lfloor x + 7 \rfloor = 4$
- B.  $\lfloor x - 2 \rfloor = 9$
- C.  $\lfloor x - 2 \rfloor + 3 = 1$
- D.  $2\lfloor x - 3 \rfloor + 1 = 7$
- E.  $2\lfloor x + 7 \rfloor + 2 = 5$

Part A

$$\lfloor x + 7 \rfloor = 4 \Rightarrow 3 < x + 7 \leq 4 \Rightarrow -4 < x \leq -3 \Rightarrow x \in (-4, -3]$$

Part B

$$\lfloor x - 2 \rfloor = 9 \Rightarrow 8 < x - 2 \leq 9 \Rightarrow 10 < x \leq 11 \Rightarrow x \in (10, 11]$$

Part C

$$\lfloor x - 2 \rfloor = -2 \Rightarrow -3 < x - 2 \leq -2 \Rightarrow -1 < x \leq 0 \Rightarrow x \in (-1, 0]$$

Part D

$$2\lfloor x - 3 \rfloor = 6 \Rightarrow \lfloor x - 3 \rfloor = 3 \Rightarrow 3 \leq x - 3 < 4 \Rightarrow 6 \leq x < 7 \Rightarrow x \in [6, 7)$$

Part E

$$2\lfloor x + 7 \rfloor = 3 \Rightarrow \lfloor x + 7 \rfloor = \frac{3}{2} \Rightarrow \text{No Solutions}$$

## C. Multiplication and Division

### Example 1.79

- A.  $\left\lfloor \frac{x}{4} \right\rfloor = 0$
- B.  $\left\lfloor \frac{x}{5} \right\rfloor = 3$
- C.  $2\left\lfloor \frac{x}{3} \right\rfloor + 4 = 10$
- D.  $\left\lfloor \frac{x}{7} \right\rfloor = 4$
- E.  $3\left\lfloor \frac{x}{5} \right\rfloor + 2 = 7$

$$\left\lfloor \frac{x}{4} \right\rfloor = 0 \Rightarrow 0 \leq \frac{x}{4} < 1 \Rightarrow 0 \leq x < 4 \Rightarrow x \in [0, 4)$$

$$\left\lfloor \frac{x}{5} \right\rfloor = 3 \Rightarrow 3 \leq \frac{x}{5} < 4 \Rightarrow 15 \leq x < 20 \Rightarrow x \in [15, 20)$$

$$2\left\lfloor \frac{x}{3} \right\rfloor = 6 \Rightarrow \left\lfloor \frac{x}{3} \right\rfloor = 3 \Rightarrow 3 \leq \frac{x}{3} < 4 \Rightarrow 9 \leq x < 12 \Rightarrow x \in [9, 12)$$

$$\left\lfloor \frac{x}{7} \right\rfloor = 4 \Rightarrow 3 < \frac{x}{7} \leq 4 \Rightarrow 21 < x \leq 28 \Rightarrow x \in (21, 28]$$

$$3\left\lfloor \frac{x}{5} \right\rfloor = 5 \Rightarrow \left\lfloor \frac{x}{5} \right\rfloor = \frac{5}{3} \Rightarrow \text{No Solutions}$$

## D. Combining Transformations

### Example 1.80

Solve:

A.  $[3x - 8] = 5$

B.  $3\left[\frac{x}{3} - 8\right] + 8 = 2$

$$5 \leq 3x - 8 < 6 \Rightarrow 13 \leq 3x < 14 \Rightarrow \frac{13}{3} \leq 3x < \frac{14}{3} \Rightarrow x \in \left[\frac{13}{3}, \frac{14}{3}\right)$$
$$\left[\frac{x}{3} - 8\right] = -\frac{6}{3} = -2 \Rightarrow -2 \leq \frac{x}{3} - 8 < -1 \Rightarrow 6 \leq \frac{x}{3} < 7 \Rightarrow 18 \leq x < 21 \Rightarrow x \in [18, 21)$$

## E. Equating Transformations

We have already seen transformations

### Example 1.81

Find the values of  $x$  that make the equality true:

$$\left[x + \frac{1}{2}\right] = \left[x + \frac{1}{3}\right] = 1$$

Equate the first and the third term:

$$\left[x + \frac{1}{2}\right] = 1 \Rightarrow 1 \leq x + \frac{1}{2} < 2 \Rightarrow \frac{1}{2} \leq x < \frac{3}{2} \Rightarrow x \in \left[\frac{1}{2}, \frac{3}{2}\right)$$

Equate the second and the third term:

$$\left[x + \frac{1}{3}\right] = 1 \Rightarrow 1 \leq x + \frac{1}{3} < 2 \Rightarrow \frac{2}{3} \leq x < \frac{5}{3} \Rightarrow x \in \left[\frac{2}{3}, \frac{5}{3}\right)$$

Now to find the value that satisfies the above, we need to find the intersection of the solution sets that we already found:

$$\left[\frac{1}{2}, \frac{3}{2}\right) \cap \left[\frac{2}{3}, \frac{5}{3}\right) = \left[\frac{2}{3}, \frac{3}{2}\right)$$

### Example 1.82

Solve

$$\left[x + \frac{1}{2}\right] = \left[x + \frac{1}{3}\right] = n$$

This will only make sense if  $n$  is an integer.

Hence, we solve for integer values of  $n$ .

Equate the first and the third term:

$$\left[x + \frac{1}{2}\right] = n \Rightarrow n \leq x + \frac{1}{2} < n + 1 \Rightarrow n - \frac{1}{2} \leq x < n + \frac{1}{2} \Rightarrow x \in \left[n - \frac{1}{2}, n + \frac{1}{2}\right)$$

Equate the second and the third term:

$$\left[x + \frac{1}{3}\right] = n \Rightarrow n \leq x + \frac{1}{3} < n + 1 \Rightarrow n - \frac{1}{3} \leq x < n + \frac{2}{3} \Rightarrow x \in \left[n - \frac{1}{3}, n + \frac{2}{3}\right)$$

Intersection of above:

$$\text{For integer value of } n: \left[n - \frac{1}{3}, n + \frac{1}{2}\right)$$

## F. Square Roots

### Example 1.83

Determine the two integers between which each square root lies.

- A.  $\sqrt{10}$
- B.  $\sqrt{27}$

$$\begin{aligned}\sqrt{9} < \sqrt{10} < \sqrt{16} &\Rightarrow 3 < \sqrt{10} < 4 \\ \sqrt{25} < \sqrt{27} < \sqrt{36} &\Rightarrow 5 < \sqrt{27} < 6\end{aligned}$$

### Example 1.84

Solve the equations below to find the range of values that satisfy the equation. Then, find the number of integers that satisfy the equation.

- A.  $\lfloor \sqrt{x} \rfloor = 1$
- B.  $\lfloor \sqrt{x} \rfloor = 12$
- C.  $\lfloor \sqrt{x} \rfloor = 4$

#### Part A

$$\begin{aligned}\lfloor \sqrt{x} \rfloor = 1 &\Rightarrow 1 \leq \sqrt{x} < 2 \Rightarrow 1 \leq x < 4 \Rightarrow x \in [1, 4) \\ &\{1, 2, 3\} \Rightarrow 3 \text{ Numbers}\end{aligned}$$

#### Part B

$$\begin{aligned}\lfloor \sqrt{x} \rfloor = 12 &\Rightarrow 12 \leq \sqrt{x} < 13 \Rightarrow 144 \leq x < 169 \\ n\{144, 145, \dots, 168\} &\Rightarrow 168 - 144 + 1 = 25\end{aligned}$$

#### Part C

We solve this using a different method

$$\begin{aligned}\lfloor \sqrt{x} \rfloor = 4 &\Rightarrow x \geq 16 \text{ and } x < 25 \Rightarrow x \in [16, 25) \\ n\{16, 17, \dots, 24\} &\Rightarrow 24 - 16 + 1 = 9\end{aligned}$$

### 1.85: Floor of a Square Root

For integers values of  $n$ :

$$\lfloor \sqrt{x} \rfloor = n \Rightarrow n^2 \leq x < (n+1)^2$$

$$\lfloor \sqrt{x} \rfloor = n \Rightarrow n \leq \sqrt{x} < n+1 \Rightarrow n^2 \leq x < (n+1)^2$$

### 1.86: Integer Solutions

The number of integer solutions of

$$\lfloor \sqrt{x} \rfloor = n, n \in \mathbb{Z}$$

Is

$$2n + 1$$

$$\lfloor \sqrt{x} \rfloor = n \Rightarrow n \leq \sqrt{x} < n+1 \Rightarrow n^2 \leq x < (n+1)^2 \Rightarrow \{n^2, n^2+1, \dots, (n+1)^2-1\}$$

The number of elements in the above is:

$$[(n+1)^2 - 1] - n^2 + 1 = [n^2 + 2n] - n^2 + 1 = 2n + 1$$

### Example 1.87

Solve the equations below to find the range of values that satisfy the equation. Then, find the number of integers that satisfy the equation.

A.  $\lceil \sqrt{x} \rceil = 5$

$$\lceil \sqrt{x} \rceil = 5 \Rightarrow x > 16 \text{ and } x \leq 25 \Rightarrow x \in (16, 25]$$
$$n\{17, 18, \dots, 25\} \Rightarrow 25 - 17 + 1 = 9$$

## G. Squares

### Example 1.88

Find the values of  $x$  that satisfy:

$$\lfloor x^2 \rfloor = 144$$

$$\lfloor x^2 \rfloor = 144 \Rightarrow 144 < x^2 < 145$$

Take Square Roots:

$$12 \leq |x| < \sqrt{145}$$

Remove the absolute value sign:

$$12 \leq x < \sqrt{145} \text{ OR } -\sqrt{145} < x \leq -12$$

$$\text{Interval Notation: } x \in (-\sqrt{145}, -12] \cup [12, \sqrt{145})$$

## H. Powers

### Example 1.89

Given that:

$$f(n) = 3^n$$

$$\text{Domain of } f = D_f = \{n \text{ such that } \lfloor n \rfloor = 3\}$$

Find the range of  $f$ .

## I. Quadratics

### Example 1.90

Find the values of  $x$  that satisfy:

$$\lfloor x^2 + 5x + 6 \rfloor = 0$$

State your answer as an exact value, not an approximation. (However, you may find  $\sqrt{5} \approx 2.23$  useful).

Using the property that  $\lfloor y \rfloor = 0 \Rightarrow 0 \leq y < 1$ , we get:

$$0 \leq x^2 + 5x + 6 < 1$$

#### First Inequality

$$0 \leq x^2 + 5x + 6$$

$$(x + 2)(x + 3) \geq 0$$

$$x \in (-\infty, -3] \cup [-2, \infty)$$

*Solution Set for First Inequality*





## Second Inequality

$$x^2 + 5x + 6 < 1 \Rightarrow x^2 + 5x + 5 < 0$$

Find the roots of the corresponding quadratic equation.

Apply the quadratic formula with  $a = 1, b = 5, c = 5$ :

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{25 - (4)(1)(5)}}{2(1)} = \frac{-5 \pm \sqrt{5}}{2}$$

$$x \in \left( \frac{-5 - \sqrt{5}}{2}, \frac{-5 + \sqrt{5}}{2} \right) \approx (-3.62, -1.39)$$

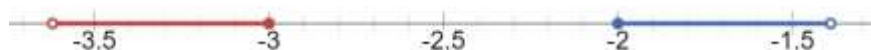
*Solution Set for  
Second Inequality*



## Merging the Two Inequalities

Our solution set must satisfy both inequalities, so we take the intersection of the two.

$$\left( \frac{-5 - \sqrt{5}}{2}, -3 \right] \cup \left[ -2, \frac{-5 + \sqrt{5}}{2} \right)$$



## Alternate Solution

For  $0 \leq k < 1$ , we must have:

$$\begin{aligned} x^2 + 5x + 6 &= k \\ x^2 + 5x + 6 - k &= 0 \end{aligned}$$

Apply the Quadratic Formula with  $a = 1, b = 5, c = 6 - k$ :

$$x = \frac{-5 \pm \sqrt{25 - (4)(1)(6 - k)}}{2} = \frac{-5 \pm \sqrt{1 + 4k}}{2}$$

$$k = 0: x = \frac{-5 + \sqrt{1}}{2} = \frac{-5 + 1}{2} \Rightarrow \{-3, -2\}$$

$$k = 1: x = \frac{-5 \pm \sqrt{5}}{2} \Rightarrow x \in \left\{ \frac{-5 - \sqrt{5}}{2}, \frac{-5 + \sqrt{5}}{2} \right\}$$

$$\left( \frac{-5 - \sqrt{5}}{2}, -3 \right] \cup \left[ -2, \frac{-5 + \sqrt{5}}{2} \right)$$

## J. Counting and Probability

### Example 1.91: Counting Integer Solutions

Find the solution set and the number of integer solutions to the following:

- A.  $\left\lfloor \frac{x}{7} + 2 \right\rfloor = 1$
- B.  $\left\lfloor \frac{x}{11} - 3 \right\rfloor = 2$
- C.  $\left\lfloor \frac{2x}{23} - \frac{3}{2} \right\rfloor = 3$

#### Part A

Convert into an inequality:

$$1 \leq \frac{x}{7} + 2 < 2$$

Subtract 2 throughout:

$$-1 \leq \frac{x}{7} < 0$$

Multiply throughout by 7:

$$-7 \leq x < 0$$

Convert to interval notation

$$x \in [-7, 0)$$

And the integer solutions from the above are:

$$\{-7, -6, \dots, -1\} \Rightarrow 7 \text{ Solutions}$$

**Part B**

$$2 \leq \frac{x}{11} - 3 < 3$$

$$5 \leq \frac{x}{11} < 6$$

$$55 \leq x < 66$$

$$x \in \{55, 56, \dots, 65\} \Rightarrow 11 \text{ Solutions}$$

**Part C**

$$3 \leq \frac{2x}{23} - \frac{3}{2} < 4$$

$$\frac{9}{2} \leq \frac{2x}{23} < \frac{11}{2}$$

$$\frac{207}{4} \leq x < \frac{253}{4}$$

$$51\frac{3}{4} \leq x < 63\frac{3}{4}$$

$$x \in \{52, 53, \dots, 63\} \Rightarrow 63 - 52 + 1 = 12 \text{ Solutions}$$

### Example 1.92: Counting Rational Solutions

Find the number of rational solutions of the form  $\frac{a}{b}$  where  $HCF(a, b) = 1$  and  $b$  is a single digit prime number.

A.  $\left\lfloor \frac{x}{7} - 2 \right\rfloor = 4$

**Solving the Inequality**

$$4 \leq \frac{x}{7} - 2 < 5$$

$$6 \leq \frac{x}{7} < 7$$

$$42 \leq \frac{x}{7} < 49$$

$$x \in [42, 49), \quad x = \frac{a}{b}, \quad HCF(a, b) = 1$$

Since  $b$  is a single digit prime number:

$$b \in \{2, 3, 5, 7\}$$

**Case I:  $b = 2$ :**

$$\left\{ \frac{84}{2}, \frac{85}{2}, \frac{86}{2}, \frac{87}{2}, \dots \right\}$$

Note that  $\frac{84}{2} = \frac{42}{1}$  and  $\frac{86}{2} = \frac{43}{1}$  do not work. If we remove the numbers that are not valid, we are left with:

$$\left\{ \frac{85}{2}, \frac{87}{2}, \dots, \frac{97}{2} \right\}$$

Note that between every two integers, there is one valid solution.

**Case II:  $b = 3$ :**

$$\left\{ 42\frac{1}{3}, 42\frac{2}{3}, 43\frac{1}{3}, 43\frac{2}{3}, \dots, 48\frac{2}{3} \right\}$$

Note that between every two integers, there are two valid solutions.

**Case II:  $b = 5$ :**

Note that between every two integers, there are four valid solutions.

**Case II:  $b = 7$ :**

Note that between every two integers, there are six valid solutions.

In total, between every two integers, there are:

$$1 + 2 + 4 + 6 = 13 \text{ Solutions}$$

**Finding the number of integer gaps**

$$(42, 43)(43, 44), \dots (48, 49)$$

The above are the seven gaps between two integers where we can have valid solutions.

The final answer is:

$$13 \times 7 = 91 \text{ Solutions}$$

### Example 1.93: Probability with Real Numbers

- Carrie picks a random real number  $x$  from 2 to 7. Find the probability that  $\lfloor x \rfloor = 3$ .
- Carrie picks a random real number  $x$  from  $a$  to  $b$  where  $a < b$ . Find the probability that  $\lfloor x \rfloor = c$ . (Assume that  $a, b$  and  $c$  are integers).
- Carrie picks a random real number  $x$  from 2.5 to 7.5. Find the probability that  $\lfloor x \rfloor = 7$ .
- Carrie picks a random real number  $x$  from 3 to 90. Find the probability that
- Carrie picks a random real number  $x$  from 2 to 10. Find the probability that  $\left\lfloor \frac{2x}{5} - \frac{2}{5} \right\rfloor = 1$ .

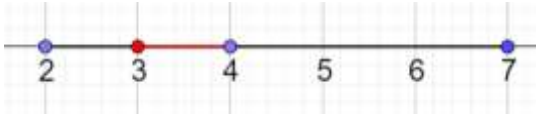
### Part A

$$3 \leq x < 4 \Leftrightarrow (3, 4]$$

$$\text{Total Length of Outcome} = 7 - 2 = 5$$

$$\text{Total Length of Successful Outcome} = 4 - 3 = 1$$

$$\text{Probability} = \frac{\text{Successful Outcome}}{\text{Total Outcome}} = \frac{1}{5}$$



### Part B

Case I:  $a \leq c < b$

$$\frac{1}{|b - a|}$$

Case II:  $c < a$  OR  $c \geq b$

$$0$$

### Part C

$$7 \leq x < 8 \Rightarrow 8 - 7 = 1$$

But from this, numbers above 7.5 are not achievable:

$$\text{Length of successful Outcome} = 7.5 - 7 = 0.5$$

$$\text{Total Length of Outcome} = 7.5 - 2.5 = 5$$

$$\text{Probability} = \frac{0.5}{5} = \frac{1}{10}$$

### Part D

$$0 \leq 3x + 1 < 1$$

$$-1 \leq 3x < 0$$

$$-\frac{1}{3} \leq x < 0$$

$$\text{Probability} = 0$$

### Part E

$$1 \leq \frac{2x}{5} - \frac{2}{5} < 2$$

$$\frac{7}{5} \leq \frac{2x}{5} < \frac{12}{5}$$

$$\frac{7}{2} \leq x < \frac{12}{2}$$

$$\text{Length of Successful Interval} = \frac{12}{2} - \frac{7}{2} = \frac{5}{2} = 2.5$$

$$\text{Length of Possible Interval} = 10 - 8 = 2$$

$$\text{Probability} = \frac{2.5}{8} = \frac{5}{16}$$

## Example 1.94: Probability with Rational Numbers

A. If I pick an integer solution at random from the solutions to  $\left\lfloor \frac{x}{22} - 1 \right\rfloor = 3$ , what is the probability that it is a prime number?

$$3 \leq \frac{x}{22} - 1 < 4$$

$$4 \leq \frac{x}{22} < 5$$

$$88 \leq x < 110$$

$$x \in \{88, 89, \dots, 109\} \Rightarrow 109 - 88 + 1 = 22$$

$$\text{Prime Number from 88 to 109 are: } \{89, 97, 101, 103, 107, 109\}$$

$$\text{Probability} = \frac{6}{22} = \frac{3}{11}$$

## 1.4 More Equations

### A. Addition and Subtraction

#### Example 1.95

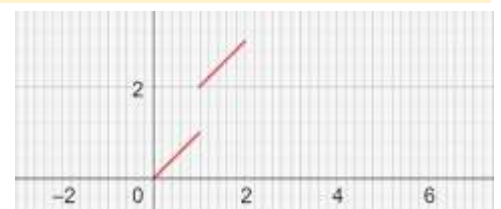
Graph

$$x + [x]$$

$$x + [x] = [x] + \{x\} + [x] = 2[x] + \{x\}$$

$$x = 1.5 \Rightarrow x + [x] = 1.5 + 1 = 2.5$$

Odd Numbers (Odd.  $x$ ) cannot be achieved



Odd Integers also cannot be achieved.

### Example 1.96

Solve

$$z + [z] = 16.2$$

Logically

$$\frac{16}{2} = 8 \Rightarrow \text{Try } 8.2 \Rightarrow 8.2 + [8.2] = 8.2 + 8 = 16.2$$

Algebraically

Write  $z = [z] + \{z\}$ , and simplify:

$$[z] + \{z\} + [z] = 16.2 \Rightarrow$$

Now, we can equate the fractional and floor parts to get:

$$2[z] = 16 \Rightarrow [z] = 8, \quad \{z\} = 0.2 \Rightarrow z = 8 + 0.2$$

### Example 1.97

Solve

$$z + [z] = 17.6$$

$$2[z] + \{z\} = 17 + 0.6$$

$$2[z] = 17 \Rightarrow [z] = 8.5 \Rightarrow \text{Contradiction} \Rightarrow z = \{\phi\}$$

### 1.98: Equation Pattern

$$z + [z] = n.m$$

Has a solution when

*Has a solution when  $n$  is even*

*Does not have a solution when  $n$  is odd*

### Example 1.99

Graph

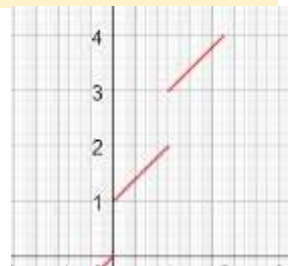
$$x + [x]$$

$$\begin{aligned} x = 1 &\Rightarrow x + [x] = 1 + 1 = 2 \\ x = 1.1 &\Rightarrow x + [x] = 1.1 + 2 = 3.1 \\ x = 1.5 &\Rightarrow x + [x] = 1.5 + 2 = 3.5 \end{aligned}$$

Even Numbers (Even.  $x$ ) cannot be achieved

Even Integers can be achieved.

Odd Integers cannot be achieved.



### Example 1.100

Solve:

A.  $z + [z] = 15.4$

B.  $z + [z] = 14.8$

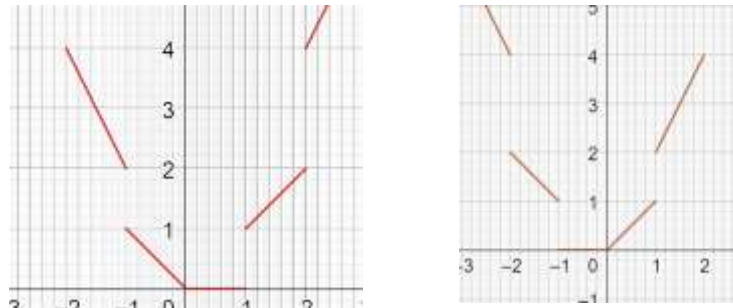
$$\begin{aligned} z + [z] = 15.4 &\Rightarrow z + [z] + 1 = 15.4 \Rightarrow z + [z] = 14.4 \\ z + [z] = 14.8 &\Rightarrow z + [z] + 1 = 14.8 \Rightarrow z + [z] = 13.8 \Rightarrow z = \{\phi\} \end{aligned}$$

## B. Multiplication and Division

### Example 1.101

Graph the functions below. Comment on any interesting properties.

- A.  $x[x]$
- B.  $x[x]$



### Example 1.102

Solve

$$x[x] = 50$$

$$7^2 = 49, 8^2 = 64 \Rightarrow 7 < x < 8 \Rightarrow [x] = 7$$

$$7x = 50 \Rightarrow x = \frac{50}{7}$$

### Example 1.103

Solve

$$x[x] = 82$$

Case I:  $x > 0$

$$\begin{aligned} x = 9 &\Rightarrow x[x] = 9[9] = 81 \\ x = 10 &\Rightarrow x[x] = 10[10] = 100 \\ x = 9.1 &\Rightarrow x[x] = 9.1[9] = 9.1(10) = 91 \\ x = 9.01 &\Rightarrow x[x] = 9.01[9] = 9.01(10) = 90.1 \end{aligned}$$

We do not get solutions from this case.

Case II:  $x < 0$

$$\begin{aligned} x = -9 &\Rightarrow x[x] = -9[-9] = 81 \\ x = -10 &\Rightarrow x[x] = -10[-10] = 100 \\ x = -9.1 &\Rightarrow -9.1[-9.1] = -9.1[-9] = 81.9 \end{aligned}$$

Solve for  $x$  with  $[x] = -9, -10 < x < -9$

$$\begin{aligned} x[x] &= 82 \\ x(-9) &= 82 \\ x &= -\frac{82}{9} = -9\frac{1}{9} \end{aligned}$$

Check the answer:

$$x[x] = \left(-\frac{82}{9}\right) \left[-9\frac{1}{9}\right] = \left(-\frac{82}{9}\right)(-9) = 82$$

## C. Number of Solutions

### Example 1.104

What is the number of nonnegative integers solution for:

$$\left\lfloor \frac{x}{5} \right\rfloor = \left\lfloor \frac{x}{7} \right\rfloor$$

PRMO

$$y = \left\lfloor \frac{x}{5} \right\rfloor = \left\lfloor \frac{x}{7} \right\rfloor$$

| $y$ | $\left\lfloor \frac{x}{5} \right\rfloor$ | $\left\lfloor \frac{x}{7} \right\rfloor$ | <i>Solution</i> | No. of Solutions |
|-----|------------------------------------------|------------------------------------------|-----------------|------------------|
| 0   | {0,1,2,3,4}                              | {0,1,2,3,4,5,6}                          | {0,1,2,3,4}     | 5                |
| 1   | {5,6,7,8,9}                              | {7,8,9,10,11,12,13}                      | {7,8,9}         | 3                |
| 2   | {10,11,12,13,14}                         | {14,15,16,17,18,19,20}                   | {14}            | 1                |
| 3   | {15,16,17,18,19}                         | {21, ...}                                | $\phi$          | 0                |

### Example 1.105

What is the number of nonnegative integer solutions for:

$$\left\lfloor \frac{x}{99} \right\rfloor = \left\lfloor \frac{x}{100} \right\rfloor$$

| $y$ | $\left\lfloor \frac{x}{99} \right\rfloor$ | $\left\lfloor \frac{x}{100} \right\rfloor$ | <i>Solution</i> | No. of Solutions |
|-----|-------------------------------------------|--------------------------------------------|-----------------|------------------|
| 0   | {0,1, ..., 98}                            | {0,1, ..., 99}                             | {0,1, ..., 98}  | 99               |
| 1   | {99, ..., 197}                            | {100, ..., 199}                            | {100, ..., 197} | 98               |
|     |                                           |                                            |                 |                  |
|     |                                           |                                            |                 |                  |

$$99 + 98 + 97 + \dots + 1 = \frac{99(98)}{2} = 4851$$

$$n + (n-1) + (n-2) + \dots + 1 = \frac{n(n-1)}{2}$$

## D. Equations

### Example 1.106

Simultaneous Equations

Quadratic Equations

## 1.5 Expressions

### A. Square Roots

### Example 1.107

Evaluate

$$\lfloor \sqrt{12} \rfloor + \lfloor \sqrt{12} \rfloor + \lfloor \sqrt{13} \rfloor + \lfloor \sqrt{13} \rfloor \dots + \lfloor \sqrt{34} \rfloor + \lfloor \sqrt{34} \rfloor$$

Collect floor terms together:

$$\underbrace{\lfloor \sqrt{12} \rfloor + \lfloor \sqrt{13} \rfloor + \dots + \lfloor \sqrt{15} \rfloor}_{\substack{\text{Each Term}=3 \\ \text{No. of Terms}=4 \\ \text{Total}=3 \times 4=12}} + \underbrace{\lfloor \sqrt{16} \rfloor + \lfloor \sqrt{17} \rfloor + \dots + \lfloor \sqrt{24} \rfloor}_{9 \text{ Terms}=4 \times 9=36} + \underbrace{\lfloor \sqrt{25} \rfloor + \lfloor \sqrt{27} \rfloor + \dots + \lfloor \sqrt{34} \rfloor}_{10 \text{ Terms}=5 \times 11=55} = \underbrace{103}_{12+36+55}$$

Similarly, evaluate all the ceiling terms together:

$$\underbrace{\lceil \sqrt{12} \rceil + \lceil \sqrt{13} \rceil + \dots + \lceil \sqrt{16} \rceil}_{5 \text{ Terms}=4 \times 5=20} + \underbrace{\lceil \sqrt{17} \rceil + \lceil \sqrt{18} \rceil + \dots + \lceil \sqrt{25} \rceil}_{9 \text{ Terms}=5 \times 9=45} + \underbrace{\lceil \sqrt{26} \rceil + \lceil \sqrt{27} \rceil + \dots + \lceil \sqrt{34} \rceil}_{9 \text{ Terms}=6 \times 9=54} = \underbrace{103}_{20+45+54}$$

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## 108 Examples